3.3 Derivatives of Inverse Functions — teacher key

Question: If you mirror a curve across y = x, what happens to its slope?

Self-paced lesson

No clock, no trimming, no skipping, no end mark, no stamp. Every tab is required in order; progress saves on the device so a student can stop and finish the next day.

Warm-up review (3 easy items)

- If $f(2) = 7$, then $f^{-1}(7)$ equals... **2.** An inverse reverses inputs and outputs. f sent 2 to 7, so f^{-1} sends 7 back to 2.
- The derivative of $y = \sin(2x)$ is... **2cos(2x).** Chain rule: the outside gives $\cos(2x)$, and the inside $2x$ brings a factor 2.
- The reciprocal of 4 is... **1/4.** To take a reciprocal, flip the number over: $4 = \frac{4}{1}$ becomes $\frac{1}{4}$.

Explore stations

Four hands-on stations (X1, X2, X3, X4). Station answers are in the teacher view of the lesson page (add `#key` to its address).

Explore: what did you see

- In Station 2, when f had slope 4, the mirror curve had slope... **1/4.** Rise and run swapped: the slope 4 over 1 became 1 over 4.
- Where f has a flat tangent, its mirror has a... **vertical tangent.** Flat means run with no rise. The mirror swaps them: rise with no run is vertical.

Mid-lesson check (Huddle tab)

- $f(x) = x^3 + x$ and $f(1) = 2$. To find $(f^{-1})'(2)$, you put into f' the number... **1.** f' is read at the input of f. Here $f(1) = 2$, so the input is $a = 1$, never the output 2.
- At some point $f'(a) = 0$. Then the inverse at the matching point has... **a vertical tangent.** Slope 0 flips to $\frac{1}{0}$, which is undefined: the mirror tangent is vertical.

Targets

- I can say how the points and slopes of f and its inverse are linked.
- I can find the slope of the inverse at a point using the derivative of f.
- I can write the tangent line to an inverse at a point.

Predictions

- The line $y = 2x$ has slope 2. Mirror it across $y = x$. The new line has slope... **1/2.** The mirror swaps x and y: $x = 2y$, so $y = \frac{1}{2}x$. Rise and run trade places.
- $f(x) = x^3$ has a flat tangent at the origin. After mirroring across $y = x$, that tangent becomes... **vertical.** Rise and run swap. A run with no rise (flat) becomes a rise with no run (vertical).

Huddle script (answers in green)

Definitions

- **Inverse:** An inverse reverses **inputs** and outputs: if f(a) = b then $f^{-1}(b) = a$.
- **Mirror:** The graph of f^{-1} is the graph of f reflected across **y = x**.
- **Slopes:** A reflection swaps rise and run, so the two slopes are **reciprocals**.

Worked example

$f(x) = x^3 + x$. Find $(f^{-1})?(2)$.

- Solve f(a) = 2. By inspection a = **1**
- The derivative is $3x^2 + 1$, so at x = 1 it is **4**
- Flip it: $(f^{-1})?(2) = 1/4$

Label the mirror

1. The curve through (1, 2) is the graph of **f**.
2. Its mirror through (2, 1) is the graph of $\mathbf{f^{-1}}$.
3. f has slope 4 at (1, 2), so f^{-1} has slope **1/4** at (2, 1).

Every level, with answers

Flip the slope

Level 1. $f(3) = 26$ and $f'(3) = 4$. Find $(f^{-1})'(26)$.
Answer: 0.25 The inverse slope is the reciprocal: $\frac{1}{4} = 0.25$.

Level 2. $f(4) = -7$ and $f'(4) = -10$. Find $(f^{-1})'(-7)$.
Answer: -0.1 The reciprocal keeps the sign: $\frac{1}{-10} = -0.1$.

Level 3. The table gives selected values of a one-to-one function f and its derivative. Find $(f^{-1})'(18)$.
Answer: 0.25 $f(2) = 18$, so use $f'(2) = 4$. The answer is $\frac{1}{4} = 0.25$.

Level 4. The table gives selected values of a one-to-one function f and its derivative. Find $(f^{-1})'(34)$.
Answer: 0.1 $f(3) = 34$, so use $f'(3) = 10$. The answer is $\frac{1}{10} = 0.1$.

Where does f? go

Level 1. $f(x) = x^3 + 2x$ and $f(1) = 3$. To find $(f^{-1})'(3)$ you evaluate f' at...
(A) x = 1 (B) x = 1/3 (C) x = 2 (D) x = 3
Answer: A $f(1) = 3$, so the input is 1. Evaluate $f'(1)$, then flip it.

Level 2. $f(x) = x^3 + x$ and $f(1) = 2$. To find $(f^{-1})'(2)$ you evaluate f' at...
(A) x = 2 **(B) x = 1** (C) x = 1/2 (D) x = 10
Answer: B $f(1) = 2$, so the input is 1. Evaluate $f'(1)$, then flip it.

Level 3. $f(x) = e^x + 3x$ and $f(0) = 1$. To find $(f^{-1})'(1)$ you evaluate f' at...
(A) x = 1 (B) x = 8 (C) x = 1/1 **(D) x = 0**
Answer: D $f(0) = 1$, so the input is 0. Evaluate $f'(0)$, then flip it.

Level 4. $f(x) = e^x + 3x$ and $f(0) = 1$. To find $(f^{-1})'(1)$ you evaluate f' at...
(A) x = 1/1 (B) x = 1 **(C) x = 0** (D) x = 8
Answer: C $f(0) = 1$, so the input is 0. Evaluate $f'(0)$, then flip it.

Slope of the inverse

Level 1. $f(x) = 5x + 4$. Find $(f^{-1})'(29)$.
Answer: 0.2 $f'(x) = 5$ everywhere, so $(f^{-1})' = \frac{1}{5} = 0.2$.

Level 2. $f(x) = x^3 + x$. Find $(f^{-1})'(2)$.
Answer: 0.25 $f(1) = 2$. $f'(1) = 4$. So $(f^{-1})'(2) = \frac{1}{4} = 0.25$.

Level 3. $f(x) = e^{3x} + x$. Find $(f^{-1})'(1)$.
Answer: 0.25 $f(0) = 1$. $f'(0) = 4$. So $(f^{-1})'(1) = \frac{1}{4} = 0.25$.

Level 4. $f(x) = e^{3x} + x$. Find $(f^{-1})'(1)$.
Answer: 0.25 $f(0) = 1$. $f'(0) = 4$. So $(f^{-1})'(1) = \frac{1}{4} = 0.25$.

Tangent to the inverse

Level 1. Write the tangent line to f^{-1} at the point $(3, 1)$.
Answer: $\frac{1}{5}(x - 3) + 1$ The point is $(3, 1)$ and the slope is $\frac{1}{5}$. So $y = \frac{1}{5}(x - (3)) + (1)$.

Level 2. Write the tangent line to f^{-1} at the point $(2, 1)$.
Answer: $\frac{1}{4}(x - 2) + 1$ The point is $(2, 1)$ and the slope is $\frac{1}{4}$. So $y = \frac{1}{4}(x - (2)) + (1)$.

Level 3. Write the tangent line to f^{-1} at the point $(1, 0)$.
Answer: $\frac{1}{4}(x - 1) + 0$ The point is $(1, 0)$ and the slope is $\frac{1}{4}$. So $y = \frac{1}{4}(x - (1)) + (0)$.

Level 4. Write the tangent line to f^{-1} at the point $(1, 0)$.
Answer: $\frac{1}{4}(x - 1) + 0$ The point is $(1, 0)$ and the slope is $\frac{1}{4}$. So $y = \frac{1}{4}(x - (1)) + (0)$.

Common misconceptions

Students say	Say instead
$(f^{-1})?$ means 1 / f?(x), so put the same x in.	Find a with f(a) = b first. Put a into f?.
f^{-1} means 1 / f.	It means the inverse function, not the reciprocal. Only the slopes are reciprocals.
The tangent to f^{-1} at (b, a) uses the point (a, b).	The inverse point has the output first: (b, a).
A flat tangent on f gives a flat tangent on f^{-1}.	Flat becomes vertical: 1/0 is undefined.